

Lab 8

2DX3

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As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is our own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario.

1. Briefly describe how the VL53L1X ToF sensor works to obtain a distance measurement

Measures how long it takes for emitted pulses of light to reach the nearest object and be reflected to the detector. The VL53L1X ToF emits pulses of infrared laser light. Then, the signal reflects from the nearest object (e.g. wall). This reflection is then collected by the ToF receiver. Finally, the time delay between the emitted and received periodic signal is measured.

The following equation is used to obtain a distance measurement:

$$\text{Measured distance} = (\text{photon travel time} / 2) \times (\text{speed of light})$$

2. What units does the VL53L1X ToF sensor return?

Returns measurements in mm (millimeters).

3. How many bits is one distance measurement for the VL53L1X ToF sensor?

16 bits (2 bytes)

4. How many data bits can be transmitted at once using UART serial communication?

8 bits (1 byte)

5. How does the UART transmitter/receiver send/receive data that is bigger than one serial package will allow?

16 bits split into two 8-bit pieces. This is known as Serialization. If data is bigger than one serial package will allow, simply just break into smaller, more manageable pieces.